monohydrate is added over a period of one hour. The precipitated crystalline *p*-toluenesulfonate salt is collected by filtration. To obtain alitame from its salt, a mixture of *Amberlite LA-1* (liquid anion exchange resin), dichloromethane, deionized water, and the salt is stirred for one hour, resulting in two clear layers. The aqueous layer is treated with carbon, clarified by filtration, and cooled to crystallize alitame.

Alternatively, tetramethylthietane amine is condensed with an *N*-protected form of D-alanine to give alanyl amide. This is then coupled to a protected analogue of L-aspartic acid to give a crude form of alitame. The crude product is then purified.

14 Safety

Alitame is a relatively new intense sweetening agent used primarily in foods and confectionary. It is generally regarded as a relatively nontoxic and nonirritant material.

Chronic animal studies in mice, rats, and dogs carried out for a minimum of 18 months at concentrations >100 mg/kg per day exhibited no toxic or carcinogenic effects. In people, no evidence of untoward effects were observed following ingestion of 15 mg/kg per day for two weeks.

Following oral administration 7–22% of alitame is unabsorbed and excreted in the feces. The remaining amount is hydrolyzed to aspartic acid and alanine amide. The aspartic acid is metabolized normally and the alanine amide excreted in the urine as a sulfoxide isomer, as the sulfone, or conjugated with glucuronic acid.

The WHO has set an acceptable daily intake of alitame at up to 0.1 mg/kg body-weight.[4]

LD_{50} (mouse, oral): >5 g/kg
LD_{50} (rabbit, skin): >2 g/kg
LD_{50} (rat, oral): >5 g/kg

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of material handled. Eye protection and gloves are recommended. Alitame should be stored in tightly closed containers, and protected from exposure to direct sunlight and higher than normal room temperatures.

16 Regulatory Status

Alitame is approved for use in food applications in a number of countries worldwide including Australia, Chile, China, Mexico, and New Zealand.

17 Related Substances

Acesulfame potassium; aspartame; saccharin; saccharin sodium; sodium cyclamate.

18 Comments

The PubChem Compound ID (CID) for alitame is 64763.

19 Specific References

1 FAO/WHO. Evaluation of certain food additives and contaminants. Fifty-ninth report of the joint FAO/WHO expert committee on food additives. *World Health Organ Tech Rep Ser* 2002; No. 913.
2 Sklavounos C. Process for preparation, isolation and purification of dipeptide sweeteners. United States Patent No. 4,375,430; 1 Mar, 1983.
3 Brennan TM, Hendrick ME. Branched amides of L-aspartyl-D-amino acid dipeptides. United States Patent No. 4,411,925; 25 Oct, 1983.
4 FAO/WHO. Evaluation of certain food additives and contaminants. Forty-sixth report of the joint FAO/WHO expert committee on food additives. *World Health Organ Tech Rep Ser* 1997; No. 868.

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Anonymous. Use of nutritive and nonnutritive sweeteners—position of ADA. *J Am Diet Assoc* 1998; **98**: 580–587.
Hendrick ME. Alitame. Nabors L, Gelardi R, eds. *Alternative Sweeteners*. New York: Marcel Dekker, 1991; 29–38.
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Renwick AG. The intake of intense sweeteners – an update review. *Food Addit Contam* 2006; **23**: 327–338.

21 Author

LY Galichet.

22 Date of Revision

5 August 2008.

Almond Oil

1 Nonproprietary Names

BP: Virgin Almond Oil
PhEur: Almond Oil, Virgin
USP-NF: Almond Oil

2 Synonyms

Almond oil, bitter; amygdalae oleum virginale; artificial almond oil; bitter almond oil; expressed almond oil; huile d'amande; oleo de amêndoas; olio di mandorla; sweet almond oil; virgin almond oil.

3 Chemical Name and CAS Registry Number

Almond oil [8007-69-0]

4 Empirical Formula and Molecular Weight

Almond oil consists chiefly of glycerides of oleic acid, with smaller amounts of linoleic and palmitic acids. The PhEur 6.0 describes almond oil as the fatty oil obtained by cold expression from the ripe seeds of *Prunus dulcis* (Miller) DA Webb var. *dulcis* or *Prunus dulcis* (Miller) DA Webb var. *amara* (DC) Buchheim or a mixture of both varieties. A suitable antioxidant may be added.

The USP32–NF27 describes almond oil as the fixed oil obtained by expression from the kernels of varieties of *Prunus dulcis* (Miller) D.A. Webb (formerly known as *Prunus amygdalus* Batsch) (Fam. Rosaceae) except for *Prunus dolcii* (Miller) D.A. Webb var. *amara* (De (Andolle) Focke).

5 Structural Formula

See Section 4.

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6 Functional Category

Emollient; oleaginous vehicle; solvent.

7 Applications in Pharmaceutical Formulation or Technology

Almond oil is used therapeutically as an emollient[1] and to soften ear wax. As a pharmaceutical excipient it is employed as a vehicle in parenteral preparations,[2] such as oily phenol injection. It is also used in nasal spray,[3] and topical preparations.[4] Almond oil is also consumed as a food substance; *see* Section 18.

8 Description

A clear, colorless, or pale-yellow colored oil with a bland, nutty taste.

9 Pharmacopeial Specifications

See Table I.

Table I: Pharmacopeial specifications for almond oil.

Test	PhEur 6.0	USP32–NF27
Identification	+	+
Absorbance	+	—
Acid value	⩽2.0	⩽0.5
Characters	+	—
Peroxide value	⩽15.0	⩽5.0
Saponification value	—	—
Specific gravity	—	0.910–0.915
Unsaponifiable matter	⩽0.9%	⩽0.9%
Composition of fatty acids	+	+
Saturated fatty acids < C_{16}	⩽0.1%	⩽0.1%
Arachidic acid	⩽0.2%	⩽0.2%
Behenic acid	⩽0.2%	⩽0.2%
Eicosenoic acid	⩽0.3%	⩽0.3%
Erucic acid	⩽0.1%	⩽0.1%
Linoleic acid	20.0–30.0%	20.0–30.0%
Linolenic acid	⩽0.4%	⩽0.4%
Margaric acid	⩽0.2%	⩽0.2%
Oleic acid	62.0–86.0%	62.0–76.0%
Palmitic acid	4.0–9.0%	4.0–9.0%
Palmitoleic acid	⩽0.8%	⩽0.8%
Stearic acid	⩽3.0%	⩽3.0%
Sterols	+	+
Δ^5-Avenasterol	⩾10.0%	⩾5.0%
Δ^7-Avenasterol	⩽3.0%	⩽3.0%
Brassicasterol	⩽0.3%	⩽0.3%
Cholesterol	⩽0.7%	⩽0.7%
Campesterol	⩽4.0%	⩽5.0%
Stigmasterol	⩽3.0%	⩽4.0%
β-Sitosterol	73.0–87.0%	73.0–87.0%
Δ^7-Stigmasterol	⩽3.0%	⩽3.0%

10 Typical Properties

Flash point 320°C
Melting point −18°C
Refractive index n_D^{40} = 1.4630–1.4650
Smoke point 220°C
Solubility Miscible with chloroform, and ether; slightly soluble in ethanol (95%).

11 Stability and Storage Conditions

Almond oil should be stored in a well-closed container in a cool, dry place away from direct sunlight and odors. It may be sterilized by heating at 150°C for 1 hour. Almond oil does not easily turn rancid.

12 Incompatibilities

—

13 Method of Manufacture

Almond oil is expressed from the seeds of the bitter or sweet almond, *Prunus dulcis (Prunus amygdalus; Amygdalus communis)* var. *amara* or var. *dulcis* (Rosaceae).[5] *See also* Section 4.

14 Safety

Almond oil is widely consumed as a food and is used both therapeutically and as an excipient in topical and parenteral pharmaceutical formulations, where it is generally regarded as a nontoxic and nonirritant material. However, there has been a single case reported of a 5-month-old child developing allergic dermatitis attributed to the application of almond oil for 2 months to the cheeks and buttocks.[6]

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of material handled.

16 Regulatory Status

Included in nonparenteral and parenteral medicines licensed in the UK. Widely used as an edible oil.

17 Related Substances

Canola oil; corn oil; cottonseed oil; peanut oil; refined almond oil; sesame oil; soybean oil.

Refined almond oil

Synonyms Amygdalae oleum raffinatum.
Comments Refined almond oil is defined in some pharmacopeias such as the PhEur 6.0. Refined almond oil is a clear, pale yellow colored oil with virtually no taste or odor. It is obtained by expression of almond seeds followed by subsequent refining. It may contain a suitable antioxidant.

18 Comments

A 100 g quantity of almond oil has a nutritional energy value of 3700 kJ (900 kcal) and contains 100 g of fat of which 28% is polyunsaturated, 64% is monounsaturated and 8% is saturated fat.

Studies have suggested that almond consumption is associated with health benefits, including a decreased risk of colon cancer.[7]

A specification for bitter almond oil is contained in the Food Chemicals Codex (FCC).[8]

19 Specific References

1 Pesko LJ. Peanut recipe softens brittle, split nails. *Am Drug* 1997; **214**(Dec): 48.
2 Van Hoogmoed LM *et al.* Ultrasonographic and histologic evaluation of medial and middle patellar ligaments in exercised horses following injection with ethanolamine oleate and 2% iodine in almond oil. *Am J Vet Res* 2002; **63**(5): 738–743.
3 Cicinelli E *et al.* Progesterone administration by nasal spray in menopausal women: comparison between two different spray formulations. *Gynecol Endocrinol* 1992; **6**(4): 247–251.
4 Christen P *et al.* Stability of prednisolone and prednisolone acetate in various vehicles used in semi-solid topical preparations. *J Clin Pharm Ther* 1990; **15**(5): 325–329.
5 Evans WC. *Trease and Evans' Pharmacognosy*, 14th edn. London: WB Saunders, 1996; 184.
6 Guillet G, Guillet M-H. Percutaneous sensitization to almond in infancy and study of ointments in 27 children with food allergy. *Allerg Immunol* 2000; **32**(8): 309–311.
7 Davis PA, Iwahashi CK. Whole almonds and almond fractions reduce aberrant crypt foci in a rat model of colon carcinogenesis. *Cancer Lett* 2001; **165**(1): 27–33.
8 *Food Chemicals Codex*, 6th edn. Bethesda, MD: United States Pharmacopeia, 2008; 38.

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20 General References

Allen LV. Oleaginous vehicles. *Int J Pharm Compound* 2000; **4**(6): 470–472.
Anonymous. Iodine 2% in oil injection. *Int J Pharm Compound* 2001; **5**(2): 131.
Brown JH *et al*. Oxidative stability of botanical emollients. *Cosmet Toilet* 1997; **112**(Jul): 87–9092, 94, 96–98.
Shaath NA, Benveniste B. Natural oil of bitter almond. *Perfum Flavor* 1991; **16**(Nov–Dec): 17, 19–24.

21 Authors

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22 Date of Revision

12 February 2009.

Alpha Tocopherol

1 Nonproprietary Names

BP: *RRR*-Alpha-Tocopherol
JP: Tocopherol
PhEur: *RRR*-α-Tocopherol
USP: Vitamin E
See also Sections 3, 9, and 17.

2 Synonyms

Copherol F1300; (±)-3,4-dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2*H*-1-benzopyran-6-ol; E307; *RRR*-α-tocopherolum; synthetic alpha tocopherol; all-*rac*-α-tocopherol; *dl*-α-tocopherol; 5,7,8-trimethyltocol.

3 Chemical Name and CAS Registry Number

(±)-(2*RS*,4′*RS*,8′*RS*)-2,5,7,8-Tetramethyl-2-(4′,8′,12′-trimethyltridecyl)-6-chromanol [10191-41-0]

Note that alpha tocopherol has three chiral centers, giving rise to eight isomeric forms. The naturally occurring form is known as *d*-alpha tocopherol or (2*R*,4′*R*,8′*R*)-alpha-tocopherol. The synthetic form, *dl*-alpha tocopherol or simply alpha tocopherol, occurs as a racemic mixture containing equimolar quantities of all the isomers.

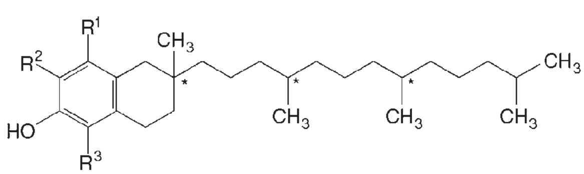
Similar considerations apply to beta, delta, and gamma tocopherol and tocopherol esters.

See Section 17 for further information.

4 Empirical Formula and Molecular Weight

$C_{29}H_{50}O_2$ 430.72

5 Structural Formula



Alpha tocopherol: $R^1 = R^2 = R^3 = CH_3$
Beta tocopherol: $R^1 = R^3 = CH_3$; $R^2 = H$
Delta tocopherol: $R^1 = CH_3$; $R^2 = R^3 = H$
Gamma tocopherol: $R^1 = R^2 = CH_3$; $R^3 = H$
* Indicates chiral centers.

6 Functional Category

Antioxidant; therapeutic agent.

7 Applications in Pharmaceutical Formulation or Technology

Alpha tocopherol is primarily recognized as a source of vitamin E, and the commercially available materials and specifications reflect this purpose. While alpha tocopherol also exhibits antioxidant properties, the beta, delta, and gamma tocopherols are considered to be more effective as antioxidants.

Alpha-tocopherol is a highly lipophilic compound, and is an excellent solvent for many poorly soluble drugs.[1–4] Of widespread regulatory acceptability, tocopherols are of value in oil- or fat-based pharmaceutical products and are normally used in the concentration range 0.001–0.05% v/v. There is frequently an optimum concentration; thus the autoxidation of linoleic acid and methyl linolenate is reduced at low concentrations of alpha tocopherol, and is accelerated by higher concentrations. Antioxidant effectiveness can be increased by the addition of oil-soluble synergists such as lecithin and ascorbyl palmitate.[4]

Alpha tocopherol may be used as an efficient plasticizer.[5] It has been used in the development of deformable liposomes as topical formulations.[6]

d-Alpha-tocopherol has also been used as a non-ionic surfactant in oral and injectable formulations.[3]

8 Description

Alpha tocopherol is a natural product. The PhEur 6.0 describes alpha-tocopherol as a clear, colorless or yellowish-brown, viscous, oily liquid. *See also* Section 17.

9 Pharmacopeial Specifications

See Table I.

Table I: Pharmacopeial specifications for alpha tocopherol.

Test	JP XV	PhEur 6.0	USP 32
Identification	+	+	+
Characters	—	+	—
Acidity	—	—	+
Optical rotation	—	+0.05° to +0.10°	+
Heavy metals	≤20 ppm	—	—
Related substances	—	+	—
Absorbance	+	—	—
at 292 nm	71.0–76.0	—	—
Refractive index	1.503–1.507	—	—
Specific gravity	0.947–0.955	—	—
Clarity and color of solution	+	—	—
Assay	96.0–102.0%	94.5–102.0%	96.0–102.0%